A SHARPENED LARGE-ENTRY MATRIX BOUND

FA/BANACH AUTOMATIC DISCOVERY RUN

Status

This is a partial result adjacent to the gap left after Theorem 2.8 of V. Muller and Y. Tomilov, *On interplay between operators, bases, and matrices*, arXiv:2010.09126. It does not fully characterize the best possible all-entry lower profile. It improves the polynomial lower exponent obtained in the source and records a basic obstruction to literally matching the source upper profile.

Source Question

The source asks whether the gap between the lower and upper estimates in its large-entry matrix theorem can be removed, at least in some sense. The relevant passage and theorem statement are shown below.

Next we consider a problem opposite in a sense to the one addressed in Theorem 2.5. For fixed operator $T \in B(H)$ we would like to find a basis $(u_n)_{n=1}^{\infty}$ giving rise to a matrix A_T of T with large entries $\langle Tu_n, u_j\rangle$. Since the task of getting the lower bounds for $\langle Tu_n, u_j\rangle$ seems to be more demanding than the one concerning the upper bounds, we restrict ourselves to the polynomial scale of bounds. However, to illustrate the sharpness of our lower estimates we provide the upper estimates as well. There is a certain gap between the two kinds of estimates, and we do not know whether the gap can be removed, at least in a sense. In terms of the size of constructed A_T, the result given below is of course weaker than Theorem 1.2. However, while in Theorem 1.2 one looks just for any operator satisfying (1.2), our theorem produces an operator having large matrix entries and belonging to the unitary orbit of a fixed $T \in B(H)$ if $W_e(T)$ is large enough.

Theorem 2.8. *Let $T \in B(H)$ be an operator which is not of the form $T = \lambda + K$ for some $\lambda \in \mathbb{C}$ and a compact operator $K \in B(H)$. Then there exist an orthonormal basis $(u_n)_{n=1}^{\infty} \subset H$ and strictly positive constants c_1, c_2, and d depending only on the diameter of $W_e(T)$ such that*

$$|\langle Tu_n, u_n\rangle| \geq d, \qquad n \in \mathbb{N}, \tag{2.9}$$

and

$$\frac{c_1 \min\{n,j\}^{1/2}}{\max\{n,j\}^{3/2}} \leq |\langle Tu_n, u_j\rangle| \leq \frac{c_2}{\max\{n,j\}^{1/2}} \tag{2.10}$$

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In the notation of the source, Theorem 2.8 gives an orthonormal basis (u_n) such that

$$|\langle Tu_n, u_n\rangle| \geq d, \qquad c_1 \frac{\min(n,j)^{1/2}}{\max(n,j)^{3/2}} \leq |\langle Tu_n, u_j\rangle| \leq \frac{c_2}{\max(n,j)^{1/2}}$$

for $n \neq j$, whenever $T \in B(H)$ is not a scalar plus compact operator.

Proof Intuition

In the source proof, the new basis vector has the form

$$u_n = (an)^{-1/2} z_n + \big(1 - (an)^{-1}\big)^{1/2} v_n.$$

The vector z_n is chosen by Ball's complex plank theorem so that it sees all previous residual vectors $(I - P_{n-1})Tu_j$ and $(I - P_{n-1})T^* u_j$ at scale $n^{-1/2}$. The source estimates those residuals by the simple lower bound $j^{1/2} n^{-1/2}$, leading to the published $j^{1/2} n^{-3/2}$ lower estimate.

The point is that the residuals decay by the product $\prod(1 - (ak)^{-1})$, not necessarily by j/n. Taking a large makes this product lose only an arbitrarily small power. This changes the lower bound from $j^{1/2} n^{-3/2}$ to $j^{\varepsilon} n^{-1-\varepsilon}$, while the same density argument still makes the orthonormal system complete.

Theorem 1 (epsilon improvement)**.** *Let H be a separable infinite-dimensional complex Hilbert space and let $T \in B(H)$ be not of the form $\lambda I + K$, where K is compact. For every $\varepsilon > 0$ there are an orthonormal basis $(u_n)_{n\geq 1}$ of H and constants $c_\varepsilon, C_\varepsilon, d > 0$ such that*

$$|\langle Tu_n, u_n\rangle| \geq d, \qquad n \geq 1,$$

and

$$c_\varepsilon \frac{\min(n,j)^{\varepsilon}}{\max(n,j)^{1+\varepsilon}} \leq |\langle Tu_n, u_j\rangle| \leq \frac{C_\varepsilon}{\max(n,j)^{1/2}}, \qquad n \neq j.$$

Proof. We follow the proof of Theorem 2.8 in the source and record only the sharpened bookkeeping. After replacing T by $T/\|T\|$, assume $\|T\| = 1$; the general case follows by rescaling the constants.

Since T is not a scalar plus compact operator, $\operatorname{diam} W_e(T) > 0$. Lemma 5.1 of the source gives constants $C, D > 0$ such that every finite-codimensional subspace $M \subset H$ contains a unit vector v with

$$|\langle Tv, v\rangle| \geq D, \quad \|Tv - \langle Tv, v\rangle v\| \geq C, \quad \|T^* v - \langle T^* v, v\rangle v\| \geq C.$$

Choose $a \geq 1$ so large that

$$\frac{4}{a} \leq D, \qquad \frac{54}{\sqrt{a}} \leq C^2, \qquad \frac{1}{a} \leq \varepsilon.$$

Run the induction from the proof of source Theorem 2.8 with this value of a. Thus $m(n)$ is defined by $n = 2^{m(n)}(2k-1)$, and at step n one chooses v_n by Lemma 5.1 and z_n by Ball's plank theorem so that, for $1 \leq j < n$,

$$|\langle (I - P_{n-1})Tu_j, z_n\rangle| \geq \frac{\|(I - P_{n-1})Tu_j\|}{2\sqrt{n}},$$

and similarly with T^*, while

$$|\langle (I-P_{n-1})y_{m(n)}, z_n\rangle| \geq 2^{-1/2}\|(I-P_{n-1})y_{m(n)}\|.$$

Set

$$u_n = \frac{1}{\sqrt{an}} z_n + \sqrt{1-\frac{1}{an}}\, v_n.$$

The source proof shows that the resulting vectors are orthonormal, satisfy the diagonal lower bound with $d = D/2$, obey the upper estimate

$$|\langle Tu_n, u_j\rangle| \leq (an)^{-1/2} \qquad (j<n),$$

and analogously for $|\langle Tu_j, u_n\rangle|$. It also proves at the creation step that

$$\|(I-P_j)Tu_j\|^2 \geq C^2/2, \qquad \|(I-P_j)T^*u_j\|^2 \geq C^2/2.$$

We sharpen the propagated residual estimate. Put

$$R_{j,k} = \|(I-P_k)Tu_j\|, \qquad R^*_{j,k} = \|(I-P_k)T^*u_j\|.$$

For $k>j$, the vector v_k is orthogonal to Tu_j, and hence

$$R^2_{j,k} = R^2_{j,k-1} - |\langle (I-P_{k-1})Tu_j, u_k\rangle|^2 \geq \left(1-\frac{1}{ak}\right) R^2_{j,k-1}.$$

The same inequality holds for $R^*_{j,k}$. Therefore, for $j<k$,

$$R^2_{j,k},\ (R^*_{j,k})^2 \geq \frac{C^2}{2}\prod_{r=j+1}^{k}\left(1-\frac{1}{ar}\right).$$

Since $ar \geq 2$ in this product, $\log(1-x) \geq -2x$ yields

$$\prod_{r=j+1}^{k}\left(1-\frac{1}{ar}\right) \geq \exp\left(-\frac{2}{a}\sum_{r=j+1}^{k}\frac{1}{r}\right) \geq \left(\frac{j}{k+1}\right)^{2/a}.$$

Now let $j<n$. Using the plank lower bound for z_n and the preceding estimate with $k=n-1$, we get

$$\begin{aligned}
|\langle Tu_j, u_n\rangle| &= \frac{1}{\sqrt{an}}\,|\langle (I-P_{n-1})Tu_j, z_n\rangle| \\
&\geq \frac{1}{\sqrt{an}}\frac{1}{2\sqrt{n}}\frac{C}{\sqrt{2}}\left(\frac{j}{n}\right)^{1/a} \\
&\geq \frac{C}{2\sqrt{2a}}\frac{j^{\varepsilon}}{n^{1+\varepsilon}},
\end{aligned}$$

because $1/a \leq \varepsilon$ and $j/n \leq 1$. The same argument with T^* gives

$$|\langle Tu_n, u_j\rangle| = |\langle T^*u_j, u_n\rangle| \geq \frac{C}{2\sqrt{2a}}\frac{j^{\varepsilon}}{n^{1+\varepsilon}}$$

for $j<n$. This is exactly the asserted two-sided off-diagonal lower bound.

Finally, the source completeness argument is unchanged: for each fixed m, the set $\{n : m(n) = m\}$ is a dyadic congruence class and

$$\sum_{n:m(n)=m} \frac{1}{2an} = \infty,$$

so the distances from the spanning subspaces to the prescribed dense sequence (y_m) tend to zero. Thus (u_n) is an orthonormal basis. $\square$

Proposition 1 (literal endpoint obstruction). *No bounded operator $T \in B(H)$, orthonormal basis (u_n), and constant $c > 0$ can satisfy*

$$|\langle Tu_n, u_j\rangle| \geq \frac{c}{\max(n,j)^{1/2}} \qquad (n \neq j)$$

for all n, j.

Proof. Fix $j = 1$. Since $Tu_1 \in H$, Parseval's identity gives

$$\sum_{n=1}^{\infty} |\langle Tu_1, u_n\rangle|^2 = \|Tu_1\|^2 < \infty.$$

The displayed lower bound would imply

$$\sum_{n=2}^{\infty} |\langle Tu_1, u_n\rangle|^2 \geq c^2 \sum_{n=2}^{\infty} \frac{1}{n} = \infty,$$

a contradiction. $\square$

Limitations and Novelty Check

Theorem 1 narrows the source's all-entry lower-bound gap but does not identify the optimal lower profile. Proposition 1 only rules out the literal profile $c/\max(n,j)^{1/2}$; it does not rule out intermediate Hilbert-matrix-type bounds such as $c/\max(n,j)$ or bounds with logarithmic corrections.

Cheap run-index search found no exact packet for arXiv:2010.09126. The local source and the later primary-source paper arXiv:2206.14266 by the same authors were checked. The later paper prescribes entries on admissible subsets under row/column summability restrictions, but the full all-entry lower-bound problem from Theorem 2.8 is not an admissible-subset instance. Web search with the phrases "On interplay between operators, bases, and matrices", "gap can be removed", "large entries", and "matrix representations" found the source paper and the later same-author array-prescription paper, but no exact settlement of this epsilon sharpening.

References

References

[1] V. Muller and Y. Tomilov, *On interplay between operators, bases, and matrices*, J. Funct. Anal. 281 (2021), no. 9, Paper No. 109158. arXiv:2010.09126.

[2] V. Muller and Y. Tomilov, *Matrix representations of arbitrary bounded operators on Hilbert spaces*, arXiv:2206.14266.

[3] K. Ball, *The complex plank problem*, Bull. London Math. Soc. 33 (2001), 433–442.

Human-Review Recommendation

Review as a likely valid partial result. The key point to check is the residual-product induction in the proof of Theorem 1; all other steps are inherited from the source proof of Theorem 2.8.